

Contact

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Top Skills

SQLAlchemy

Pydantic

FastAPI

Languages

Latin (Elementary)

Romanian

English

French

Certifications

HashiCorp Vault Enterprise

AWS Fundamentals Specialization

Artificial Intelligence Foundations:
Thinking Machines

Artificial Intelligence Foundations:
Neural Networks

Spring: Spring MVC

Publications

GitHub Repos

GitLab Repos

Coding Best Practices for Building
Intelligent B2C Cloud Native
Applications

Ovidiu Mura

Senior Software Engineer | AI Solutions | Python

Portland, Oregon Metropolitan Area

Summary

Experienced Software Engineer with over 15 years of industry experience spanning domain knowledge in Financial Delivery Systems, Healthcare Integrated Systems, Transportation, eCommerce applications, Automation and Firmware frameworks, and Computer Security.

As a Software Engineer, I apply my expertise in cloud, blockchain, and cryptography to develop secure and scalable applications for data annotation and machine learning. I have over four years of experience in designing and implementing cybersecurity solutions for various domains, including financial, healthcare, and e-commerce.

I have a strong background in full-stack development, using Java, C#, Python, NodeJS, AngularJS, and other technologies to create user-friendly and high-performance web applications and APIs. I also have a graduate certificate and a master's degree in computer security from Portland State University, where I learned the latest techniques and best practices in cryptography, network security, and data privacy. I am passionate about solving complex and challenging problems that have a positive impact on society and the environment.

Experience

TEKsystems

Software Engineer

March 2026 - August 2026 (6 months)

- Design and develop scalable, cloud-native software solutions using Python, C#, JavaScript/TypeScript, and distributed systems architectures.
- Develop and integrate AI-powered tools and agentic workflows to support software engineering, automation, and technical problem solving.

- Build and integrate Model Context Protocol (MCP) servers and tools, enabling AI agents to interact with external services, APIs, and technical environments.
- Containerize and deploy AI and backend services using Docker, Kubernetes, and Google Kubernetes Engine (GKE).
- Design and implement REST APIs, backend services, and distributed application components.
- Troubleshoot complex issues across application code, APIs, databases, messaging systems, and cloud infrastructure.
- Apply event-driven and asynchronous architectures to build scalable and resilient distributed systems.
- Work with cloud services and infrastructure across AWS, GCP, and Kubernetes-based environments.
- Collaborate with engineering teams to analyze requirements, evaluate technical approaches, and implement maintainable solutions.
- Leverage LangChain, LangGraph, Google ADK, MCP, and LLM technologies to develop and evaluate AI-enabled engineering workflows.
- Contribute to system design, code reviews, debugging, performance optimization, and continuous improvement of production software.

US Tech Solutions

Software Engineer

April 2025 - February 2026 (11 months)

United States

I played a pivotal role in developing AI solutions for e-commerce focusing on scalability and reliability.

- Designed and implemented MCP server tools deployed in Docker and Kubernetes environments.
- Utilized Python to implement AI systems, enhancing the overall performance of applications.
- Engaged in building and training ML models for Gen AI, optimizing workflows and integrations.

Highlight

System Developer / Full Stack Developer

March 2024 - March 2025 (1 year 1 month)

Portland, OR

System Developer supporting SBA IT service center, contributing to system architecture, automation, API integrations, and cloud-based solutions in a

hybrid federal environment. Solutions deployed on AWS and Azure Cloud Platforms. Led the migration of on-prem Azure DevOps to Azure Cloud, improving scalability, security, and CI/CD reliability.

DataAnnotation

Software Engineer

January 2024 - February 2024 (2 months)

United States

- Provide diverse problems and solutions for a coding chatbot
- Write high-quality answers and code snippets
- Evaluate code quality produced by AI models for correctness and performance
- Reviewing scripts written in programming language such as Python, JavaScript, HTML, C++, C# and SQL

Apex Systems / Bank of America

Full-Stack Software Developer / Application Architect

December 2021 - December 2023 (2 years 1 month)

I designed and implemented Cybersecurity solutions for applications

I wrote documentation and configured systems to comply with Cybersecurity standards

I completed the certification course for HashiCorp Vault

I configured and wrote best practice documentation for HashiCorp Vault, Cyberark and gMSA

I developed Java, C# and Python programs to authenticate and rotate passwords for applications

I configured Active Directory, Oracle, SQL Server service accounts to rotate passwords automatically and store them in HashiCorp Vault using REST APIs

Pactera EDGE

Software Engineer

January 2020 - November 2021 (1 year 11 months)

I developed a Elliptic Curve Digital Signature Algorithm for NodeJS application to secure transactions on markaaz.com hosted in the cloud, markaaz.com

I developed front-end User Info page (AngularJS) and back-end (NodeJS) APIs for markaaz.com

I developed Blockchain application for digital identity using Ethereum ledger

I developed back-end microservices - REST APIs for B2C cloud application

I developed Java applications processing large repositories of metadata

I developed Python, and PowerShell scripts to automate the process of loading metadata into existing and new repositories

I wrote store procedures to insert, delete, update data in SQL server and I developed Java programs to use these store procedures processing metadata

I setup FTP server over SSL connection to transfer files from one server to the other

I developed Java program which generates Metadata and Models for multiple types of entities, attributes, application types, type lists, list items and their relationships

I use Go to build and deployment for Cloud Native applications on Azure

I developed and deployed accelerators, data pipelines and machine learning pipelines using Python framework Kedro, Data Validator Tool, Kedro Streaming, Spark, TensorFlow

I developed programs to load data into Data Lake, Blob, S3, BigQuery

I developed programs to convert Databricks pipelines to Synapses pipelines

I developed program to parse SSIS (SQL Server Integration Services) packages and extract the SQL Tasks and Process Tasks, then I built a csv report file presenting the details for the SQL tasks and the flow of the execution

Completed Machine Learning, Neural Networks and Deep Learning training courses and built models to predict products prices

I designed and implemented Java REST APIs using Servlets and Spring Boot

Self

Full Stack Engineer

September 2019 - July 2020 (11 months)

Portland, Oregon, United States

I developed a Market Hub web application in NodeJs, React, JavaScript, HTML, CSS, Express, axios, and deployed on Google Cloud Firebase environment

The application built for Chrome Version 83, Windows

Url: market-hub-45641 [dot] firebaseapp [dot] com

Infosys

Technology Lead

April 2018 - December 2019 (1 year 9 months)

Portland, Oregon Area

- Adding value to Software Development life-cycle stages

- Create detailed design artifacts like program specifications, test plans
- Independently develop and review code and contribute to the go-live plan
- Fixing defects in production for e-commerce web applications and mobile
- Deploy/configure/develop/support features in Salesforce.com application
- Building pages for e-commerce applications and configure feeds properly
- Writing Python scripts to analyze Big Data and generating reports

UST Global

Software Engineer

June 2017 - March 2018 (10 months)

Hillsboro OR

- BIOS Automation Developer
- Python Developer
- Applying Behavior Driven Development
- C++ Developer
- Implementing BIOS Automation test cases using Python, and Gherkin
- Participate in daily stand-up meeting and reporting the work status
- Implement unit tests with Python for BIOS Automation
- Development tools used: PyCharm, Perforce, Smart Bear – Collaborator, Jama, TTK, Vetra Box
- Python Developer writing automation on linux, Fedora, Windows
- Writing REST api for failure analyzer tool
- Develop failure analyzer tools using Object Oriented Design and Python
- Develop tools to report results for testing software and analyze the failures
- Developing automation for Windows 10 and IoT and running the tests on TWS testing engine

Compunnel Staffing

Engineer: System Validation

January 2017 - June 2017 (6 months)

Hillsboro, OR

- Implementing WinIoT automation framework in PowerShell, Python for Joule 570x, WinIoT platform.
- Installing drivers and Firmware on Joule 570x devices
- Troubleshooting drivers and firmware
- Installing Operating System on Joule 570x
- Writing automation platform on Joule 570x using Powershell and Python technology
- Writing programs to test memory, cpu and disk using C, and C++
- Installing Windows Server 2016 on Testing Lab Network

- Installing Hyper-V virtual machines on Windows Server 2016
- Installing Windows 10 and Ubuntu on virtual machines
- Creating Test environment on the server and virtual machines to be able to run automation scripts written in PowerShell, Python, C, C++ and C#
- Writing Universal Applications on WinIoT platform
- Writing scripts to measure the performance of WinIoT platform operations
- Writing reports to reflect the reliability and stability of the Joule 570x devices

Ampcus Inc

Software Engineer

April 2016 - December 2016 (9 months)

Hillsboro, OR

- Installing drivers and firmware in test target with Intel processors, PCH, Beagle I2C, Aardvark devices, eight relays [Power Button, Reset Button, Clear CMOS, ME Recovery, FOJ Button, Manufacture Relay, Chipset Reset, Flash Drive Relay], Saleae Logic Analyzer Device, Monitor Serial Port, Keyboard Serial Port
- Running/Troubleshooting C# test scripts in Cloud application
- Running/Troubleshooting Python test scripts in Cloud application
- Installing/troubleshooting Wind River Simics
- Running test scripts with Wind River Simics
- Configure IPMI, IPC and Diagnostic bus interfaces for Intel hardware

TEKsystems

Software Engineer

February 2015 - March 2016 (1 year 2 months)

Beaverton, OR

- Daily stand-up meeting reporting progress on the automation and performance testing for nike.net
- Writing UI Automation scripts to cover the nike.net application
- Writing Performance Testing scripts to measure the nike.net application performance respond time
- Writing java application tool for monitoring the resources on the nike.net servers
- Releasing daily reports for ui automation and performance suites
- Leading the UI Automation and Performance testing for nike.net application which places about \$13 billions orders for 2015
- Sprint release every 2 weeks
- Groovy, Spock, Geb, gradle, Saucelabs, jMeter, fasterxml/json, IntelliJ IDEA, Version One

McKesson Health Solutions

Software Engineer

January 2012 - February 2015 (3 years 2 months)

Beaverton, OR

- Technical Writing
- Create Testing Strategy Documents
- Standard operating Procedures Documents
- Reviewing Requirements Documents
- Create Reports Documents
- Write Test Scenario and Test Cases
- Create Test Strategy for Java Client-Server Application - Contract manager
- Written Java Selenium/Fluentlenium Scripts; QTP scripts with Visual Basic Script; soapUI scripts; web services test scripts in C#
- Configure/Deploy Client-Server Java Applications
- Setting Complex Configuration of Single Sign on using SAML 2.0 on OpenAm, OpenIdp, WSO 4.0
- Troubleshoot single Sign On using Spnego
- Maintaining Virtual Windows and linux Machines
- Leading and designing the Regression Testing Automation for McKesson Contract Manager application

Fiserv

Software Engineer

February 2010 - January 2012 (2 years)

Corvallis, Oregon, United States

- Network Deploying build versions on Tomcat and Glassfish, Linux platform – SUSE, RHEL
- Designing, writing, and executing test cases in XML files as an input for TAF (Test Automation Framework, specific to Fiserv's Acumen product), an engine using the application's API written in Java.
- Actively participating as Software Developer Engineer (Test) in Life Cycle of Acumen – Financial Delivery
- Software implementing SCRUM – Agile Methodology
- Write Test Scenarios in C# for web services testing jobs run successfully on Batch Processing server
- Testing Encryption Algorithms
- Install SUSE, RHEL, Oracle 11g, Microsoft SQL Server on physical and virtual machines
- Written SQL Queries using Oracle
- Maintaining code using Eclipse, Visual Studio, and NetBeans

- Test automatically generated pdf documents in Credit Union Bank - Financial Delivering System

Education

Portland State University

Graduate Certificate in Computer Security, Computer Science · (2018 - 2020)

Portland State University

Master's degree, Computer Science · (2017 - 2020)

Portland State University

Bachelor of Science, Computer Science · (2006 - 2009)